Quantum Machine Learning

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What is Quantum Machine Learning?

Quantum Machine Learning (QML) uses trainable quantum circuits to learn patterns from data, often via **Quantum Neural Networks** (QNNs). Unlike QNNs, classical feed-forward networks process input $\mathbf{x}$ through layers using linear weights W_i and non-linear activation functions σ_i, where each layer outputs $\mathbf{z} = \sigma_i(W_i^T \mathbf{x})$.

VQC Architecture

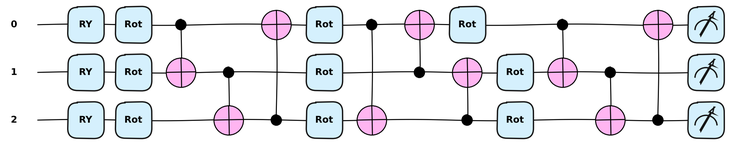
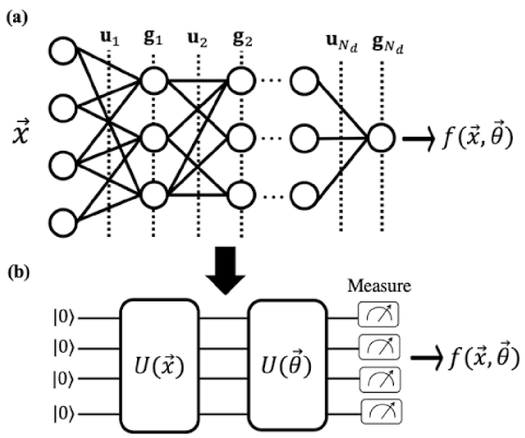


Figure 1. The 3 layer 3 qubit VQC with fixed state preparation

The circuit shown above is an example of a VQC circuit.
The output of the circuit is the energy (Hamiltonian) of the system defined by:

$$H = 1.0(Z_0 \otimes Z_1) + 0.5X_2$$

Just like classical nets, QNNs can have different architectures. The architecture we focus on due to simplicity is the **Variational Quantum Circuit** (VQC) approach. In this design, qubits are prepared then rotated and entangled through several layers. Entangling allows the network to learn deep relationships between qubits, functioning like σ_i in classical nets.



Optimization

Training quantum circuits, like classical networks, requires minimizing a cost function: $\arg\min_{\theta \in \mathbb{R}^n} f(\theta)$. While classical optimizers (e.g., SGD, Adam) rely on the gradient ∇f, adapting them to quantum systems introduces key challenges:

- Due to quantum mechanics, gradients must be calculated using the phase shift rule which requires evaluating the circuit twice per parameter
- An n qubit state belongs to a 2^n dimensional space leading to massive regions where the cost function is approximately flat (**barren plateaus** [2]).

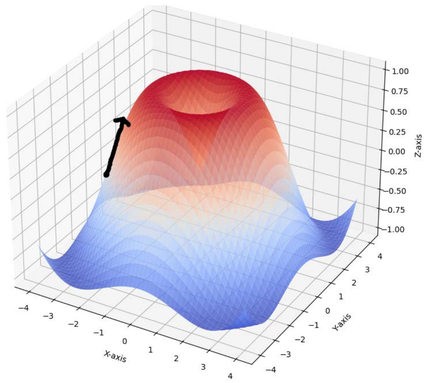


Figure 3. The gradient vector pointing in the direction of steepest ascent

Strategies for Optimization

Methods are divided between gradient and gradient-free, the former relies on variations of the update the rule for parameters θ_t:

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$$

Quantum Natural Gradient

While gradient descent searches euclidean space, **quantum natural gradient** (QNG) searches information space [3]. This is done by adjusting the gradient by some factor to account for its effect on the final distribution. In QNG, we use the inverse of the Fubini-Study Metric, g:

$$\theta_{t+1} = \theta_t - \eta g^{+}(\theta_t) \nabla f(\theta_t)$$

Benchmarking Approaches

We benchmarked several gradient and gradient-free approaches by using them to find the ground state of the quantum system described by the circuit in Figure 1:

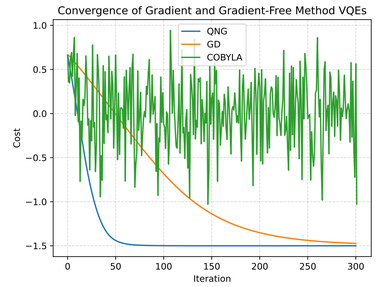


Figure 4. A graph of cost over epoch of the VQC circuit. The green line is the convergence line of COBYLA, blue is simple gradient descent, and orange is natural gradient descent

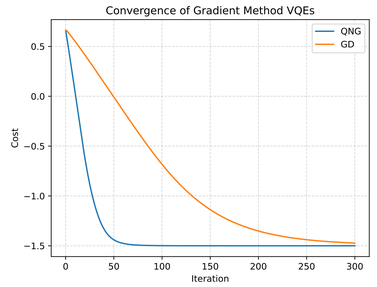


Figure 5. The same cost graph as to the left but with COBYLA removed for clarity

Applications of Quantum Machine Learning

- Native high dimensional state space makes representing high dimensional data natural for quantum computers
- Quantum algorithms consistently demonstrate lower parameter counts compared to classical models
- Quantum models have proved capable of finding expressive patterns within very small datasets without overfitting [1]

References

[1] Shreeya Sanjeev Gokhale, Rishit Ravi Dhote, and Radhakrishnan Delhibabu. A review of quantum machine learning algorithms, applications, and emerging advantages. *Discover Computing*, 29(1):226, apr 2026.

[2] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, nov 2018.

[3] James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020.